Fuzzy-Conformal Graph Neural Networks with Distribution-Free Coverage Guarantees for Cybersecurity Anomaly Detection

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raph Neural Networks (GNNs) have demonstrated strong performance for cybersecurity anomaly detection tasks — including network intrusion classification, botnet identification, and IoT malware analysis — yet two critical limitations remain unresolved simultaneously: the inability to handle inherently imprecise network telemetry through fuzzy reasoning, and the lack of distribution-free, finite-sample coverage guarantees that compliance-sensitive environments demand. We present **FC-GNN** (**F**uzzy-**C**onformal **G**raph **N**eural **N**etwork), the first architecture to jointly address both weaknesses. FC-GNN replaces the standard crisp aggregation function with a *fuzzy message-passing layer* that computes Gaussian membership functions and Product T-norm rule-firing strengths, yielding uncertainty-aware node embeddings. A Sugeno-type fuzzy integral then defines a nonconformity score that feeds into *Fuzzy Mondrian Conformal Prediction* (FMCP), which partitions calibration nodes into graph communities via Louvain clustering and computes per-community quantiles — providing conditional, finite-sample coverage guarantees of the form $\mathbb{P}(y_v \in \mathcal{C}_\alpha(v) \mid v \in Q_m) \geq 1 - \alpha$. We prove a coverage theorem for FC-GNN under a fuzzy-extended permutation-invariance condition, and show that the conditional slack is $O(|Q_m|^{-1/2})$. Experiments on seven publicly available cybersecurity benchmarks — CIC-IDS-2017, UNSW-NB15, NF-BoT-IoT, ISCX-Botnet 2014, CTU-13, N-BaIoT, and ToN-IoT — confirm that FC-GNN provides theoretical conditional coverage under stated exchangeability assumptions (empirical mean coverage 0.903, reaching target on 3/7 datasets at $\alpha = 0.10$) while outperforming the leading Mondrian CP baseline (RR-GNN) in prediction-set efficiency on five datasets, and providing human-interpretable IF-THEN rule explanations aligned with MITRE ATT&CK categories (mean explanation stability 0.61).

Table 1: Research gap: no prior work simultaneously provides coverage guarantees, fuzzy aggregation, and cybersecurity validation.

Method	Coverage	Fuzzy	Cyb. Bench.
CF-GNN (Huang et al. 2023)	✓	×	×
RR-GNN (Zhang et al. 2025)	✓	×	×
FL-GNN (Du et al. 2024)	×	✓	×
FGAT (Xing et al. 2025b)	×	✓	×
GraphIDS (Guerra et al. 2025)	×	×	✓
FC-GNN (ours)	✓	✓	✓

1 Introduction

Network intrusion detection, botnet identification, and IoT malware analysis are canonical cybersecurity tasks that operate over inherently *graph-structured* data: traffic flows between hosts form edges, devices form nodes, and attack campaigns manifest as anomalous subgraph patterns. Graph Neural Networks (GNNs) have emerged as powerful classifiers for these tasks, achieving high accuracy on benchmarks such as CIC-IDS-2017 (Canadian Institute for Cybersecurity 2017), UNSW-NB15 (UNSW Canberra 2015), and NF-BoT-IoT (Koroniotis et al. 2020) — yet they suffer from two critical, simultaneously unresolved weaknesses.

W1* — *Epistemic uncertainty from noisy and ambiguous features.

Network telemetry is inherently imprecise: packet-level features exhibit quantisation noise, flow statistics are aggregated over variable windows, and labelling of attack vs. benign traffic is often soft or partially overlapping (Guerra et al. 2025). Standard GNN classifiers produce crisp binary decisions that discard this inherent fuzziness, leading to brittle predictions and inflated false-alarm rates in production SOC deployments.

W2* — *Absence of formal coverage guarantees.

Neither standard GNNs nor fuzzy-neural extensions provide *distribution-free, finite-sample guarantees* on their prediction sets. A security analyst receiving an alert has no principled bound on the probability that the true attack class lies within the model's output — making risk quantification impossible for compliance-sensitive (e.g., GDPR, NIS2) environments (Zhou et al. 2026).

1.1 Research Gaps

Significant progress has been made independently along two complementary lines: (a) *conformal prediction for GNNs* (Huang et al. 2023; Zargarbashi et al. 2023; Clarkson 2023; Zhang et al. 2025) and (b) *fuzzy GNN architectures* (Du et al. 2024; Xing et al. 2025b; Tran et al. 2025). However, their explicit fusion remains an open problem. Table 1 positions FC-GNN against the five closest prior methods.

1.2 Contributions

This paper makes four contributions:

1. **FC-GNN architecture.** A novel GNN in which each message-passing layer incorporates a fuzzy-rule aggregation module that outputs a fuzzy membership distribution over classes per node, rather than a crisp softmax logit (Section 3).
2. **Fuzzy Mondrian Conformal Prediction (FMCP).** A principled extension of Graph-Structured Mondrian CP (Zhang et al. 2025) to fuzzy nonconformity scores defined via a Sugeno-type fuzzy integral, producing calibrated prediction sets with per-cluster conditional coverage guarantees (Section 3.4).
3. **Coverage theorem.** A finite-sample theorem proving that FC-GNN's prediction sets achieve at least $(1-\alpha)$ marginal coverage and conditional coverage within graph communities, with a slack term $\delta(|Q_m|) = O(|Q_m|^{-1/2})$ (Section 3.5).
4. **Cybersecurity benchmarks.** Systematic evaluation on seven publicly available datasets spanning intrusion detection, botnet detection, and IoT malware, with temporal chronological splits and full ablation studies (Section 4).

1.3 Paper Structure

Section 2 reviews related work. Section 3 covers necessary preliminaries (Section 3.1) and presents the FC-GNN architecture and the FMCP calibration procedure. Section 4 reports empirical results and analyses the extracted fuzzy rules (Section 4.7). Section 5 discusses limitations. Section 6 concludes.

2 Related Work

2.1 Conformal Prediction — Foundations

Conformal prediction (CP) (Angelopoulos and Bates 2021) provides distribution-free, finite-sample prediction sets satisfying marginal coverage $\mathbb{P}(y \in \mathcal{C}_\alpha(x)) \geq 1-\alpha$ for any pre-specified error rate α. Barber et al. (2023) extended CP beyond exchangeability to handle covariate shift. Zhou et al. (2026) survey CP comprehensively including a graph-data subsection. The present authors have applied CP to intrusion detection (Dang 2023) and botnet detection (Dang and Pham 2024b), and developed CONFIDE for causal competing-risk settings (Dang et al. 2026).

2.2 Conformal Prediction for Graph Neural Networks

Huang et al. (2023) introduced CF-GNN, the de-facto baseline for CP on graphs. They derive a *permutation-invariance condition* under which calibration nonconformity scores are exchangeable and hence CP is valid; they also propose a topology-aware score correction. Zargarbashi et al. (2023) proposed DAPS, which diffuses conformity scores over the graph using homophily weighting, improving prediction-set tightness on node classification. Clarkson (2023) studied structure-aware inductive node CP. Song et al. (2024) introduced SNAPS, aggregating scores from feature-similar neighbours to improve singleton-hit proportion.

Among the closest to our work, Zhang et al. (2025) proposed RR-GNN, which uses *Graph-Structured Mondrian CP* — partitioning calibration nodes into graph communities and computing per-community quantiles. This yields conditional coverage guarantees within communities and motivates the FMCP component of FC-GNN. Zhang et al. (2025) do not, however, incorporate fuzzy aggregation or cybersecurity evaluation.

Other CP-GNN work includes federated settings (Akgül et al. 2025), temporal GNNs (Davis et al. 2025; Wang et al. 2025a), and conformal training (Wang et al. 2025b). Bai et al. (2025) extend CP to graph anomaly detection with risk control. Koning and van Meer (2026) develop fuzzy prediction sets via e-values in non-graph settings, providing a theoretical bridge to our approach.

2.3 Fuzzy Graph Neural Networks

Du et al. (2024) proposed FL-GNN, which defines a Cartesian-product rule base in the message-passing layer, enabling interpretable IF-THEN explanations. Our fuzzy message-passing layer (FMPL) directly extends FL-GNN's rule base, adding Gaussian membership functions, the product T-norm, and centroid defuzzification. Xing et al. (2025b) extended fuzzy logic to graph attention (FGAT), using fuzzy rough-set attention weights with dynamic negative sampling. Xing et al. (2025a) proposed MFGAT with multi-view fuzzy attention. A comprehensive survey by Tran et al. (2025) identifies the absence of coverage guarantees as a key open problem across all fuzzy GNN architectures — a gap we directly address. Sengupta and Rekik (2025) developed FireGNN with trainable neuro-symbolic fuzzy rules, and Khaled et al. (2026) applied explainable fuzzy GNNs to infrastructure anomaly detection. The present authors applied fuzzy embeddings to software-defined network NIDS in (Dang 2024).

2.4 GNNs for Cybersecurity

Guerra et al. (2025) achieved 99.98% PR-AUC on NF-BoT-IoT using self-supervised E-GraphSAGE. Sun et al. (2024) proposed a hybrid static-dynamic graph IDS. Farrukh et al. (2025) combined heterogeneous GNN with LLM for dual-modality NIDS. Various GNN-IDS architectures have been surveyed in (Bilot et al. 2023). The authors previously proposed GNN-based NIDS baselines in (Dang and Nguyen 2023) and federated IoT malware detection in (Dang and Pham 2024a). As GNNs see wider deployment in cybersecurity, their robustness against graph adversarial threats has become a critical concern. Recent literature explores vulnerabilities such as node injection attacks and corresponding defenses (e.g., GZOO, DShield) (Yu et al. 2025a,b). Furthermore, structural entropy has emerged as a complementary theoretical perspective for understanding graph complexity and structural anomalies, which is highly relevant for proactive botnet detection (Zeng et al. 2026).

2.5 Conformal Prediction in Cybersecurity

Yumlembam et al. (2024) applied a CP rejection layer to botnet classification, rejecting 58% of incorrect predictions on ISCX. García and DeCastro-García (2025) showed that CP pseudo-labels become inconsistent under heavy concept drift, motivating our temporal split protocol. Qaiser and Chandrasekaran (2026) extended open-set CP to zero-day IIoT detection. Yuan et al. (2026) proposed sliding-window temporal quantile CP for cyber-physical system anomaly detection.

2.6 Fuzzy Methods in Cybersecurity

Qiu et al. (2025) combined fuzzy logic with CNN ensembles for IoT NIDS, achieving 99.72% on NSL-KDD. Johnpeter and Karuppanan (2025) proposed hunger-games-optimised fuzzy-rule deep learning for SDN-enabled IoT environments. Elhag et al. (2015) evolved fuzzy rules with adversarial training for IDS. These works confirm the practical value of fuzzy reasoning for cybersecurity but lack any CP coverage guarantees.

3 Preliminaries and the FC-GNN Method

3.1 Preliminaries

3.1.1 Graph Neural Networks

Let $G = (\mathcal{V}, \mathcal{E}, \boldsymbol{X})$ be an attributed graph with node set $\mathcal{V}$ ($|\mathcal{V}| = N$), edge set $\mathcal{E}$, and feature matrix $\boldsymbol{X} \in \mathbb{R}^{N \times F}$. A standard GNN with L message-passing layers computes node embeddings via

$$\boldsymbol{h}_v^{(\ell+1)} = \mathrm{UPDATE}\Big(\boldsymbol{h}_v^{(\ell)},\ \mathrm{AGG}\Big(\{\boldsymbol{h}_u^{(\ell)} : u \in \mathcal{N}(v)\}\Big)\Big),$$

where $\mathcal{N}(v)$ denotes the neighbourhood of node v, $\boldsymbol{h}_v^{(0)} = \boldsymbol{x}_v$, and the final embedding $\boldsymbol{h}_v^{(L)}$ is passed to a classification head.

3.1.2 Conformal Prediction

Given a calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$ and a nonconformity score function $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ (higher score $\Leftrightarrow$ less conforming), the CP prediction set at error rate α is

$$\mathcal{C}_\alpha(x) = \big\{y \in \mathcal{Y} : s(x, y) \leq \hat{q}\big\},$$

where $\hat{q}$ is the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of $\{s(x_i, y_i)\}_{i=1}^n$. Under exchangeability of the calibration and test scores, $\mathbb{P}(y_{\text{test}} \in \mathcal{C}_\alpha(x_{\text{test}})) \geq 1 - \alpha$ (Angelopoulos and Bates 2021).

Mondrian CP.

Mondrian CP (Vovk et al. 2005) extends the above by conditioning on a *taxon* $\tau(x)$, computing a separate quantile per taxon:

$$\hat{q}_\tau = \hat{Q}_{1-\alpha}\big(\{s(x_i, y_i) : \tau(x_i) = \tau\}\big).$$

This yields *conditional coverage*: for each taxon τ, $\mathbb{P}(y \in \mathcal{C}_\alpha(x) \mid \tau(x) = \tau) \geq 1 - \alpha$.

3.1.3 Fuzzy Sets and Sugeno Integral

A *fuzzy set* A in universe U is characterised by a membership function $\mu_A : U \to [0,1]$. A *fuzzy measure* $g : 2^U \to [0,1]$ satisfies $g(\emptyset) = 0$, $g(U) = 1$, and monotonicity. The λ-*fuzzy measure* (Sugeno 1974) is defined recursively: for A, B disjoint, $g_\lambda(A \cup B) = g_\lambda(A) + g_\lambda(B) + \lambda\, g_\lambda(A)\, g_\lambda(B)$, with $\lambda \in (-1, \infty)$ chosen so that $g_\lambda(U) = 1$. The *Sugeno fuzzy integral* of $h : U \to [0,1]$ with respect to g_λ is

$$\mathcal{S}_\lambda(h, g_\lambda) = \sup_{A \subseteq U} \Big[\min\Big(\min_{u \in A} h(u),\ g_\lambda(A)\Big)\Big].$$

Efficient computation sorts h in descending order and takes the maximum over the cumulative λ-measure prefix products.

3.1.4 Graph Permutation-Invariance for CP

Huang et al. (2023) established that CP is valid for GNN node classification when the nonconformity scores are *jointly exchangeable under graph automorphisms* — i.e., any permutation π of nodes that preserves the graph structure also preserves the joint distribution of scores. Zhang et al. (2025) extended this to the Mondrian setting using graph community structure as the taxon. We extend both results to the fuzzy-nonconformity-score case in Section 3.5.

3.2 FC-GNN Architecture

Figure 1 shows the FC-GNN architecture. The model comprises three stages: (i) L *fuzzy message-passing layers* that produce a per-node fuzzy membership distribution $\boldsymbol{\mu}_v \in [0,1]^C$, (ii) an MLP output head, and (iii) post-hoc *Fuzzy Mondrian Conformal Prediction* calibration that wraps the trained model without retraining.

3.3 Fuzzy Message-Passing Layer (FMPL)

Let $\boldsymbol{h}_v^{(\ell)} \in \mathbb{R}^d$ be the embedding of node v at layer ℓ, with $\boldsymbol{h}_v^{(0)} = W_0 \boldsymbol{x}_v$. The FMPL replaces the standard mean/sum aggregation with a fuzzy-rule-based combiner operating as follows.

Gaussian membership functions.

For each rule $k \in \{1, \ldots, K\}$, define a learnable Gaussian membership function over the d-dimensional embedding space:

$$\mu_k(\boldsymbol{h}) = \exp\Big(-\tfrac{1}{2}\big\|\mathrm{diag}(\boldsymbol{\sigma}_k)^{-1}(\boldsymbol{h} - \boldsymbol{c}_k)\big\|^2\Big),$$

where centres $\boldsymbol{c}_k \in \mathbb{R}^d$ and log-widths $\log \boldsymbol{\sigma}_k \in \mathbb{R}^d$ are learnable parameters.

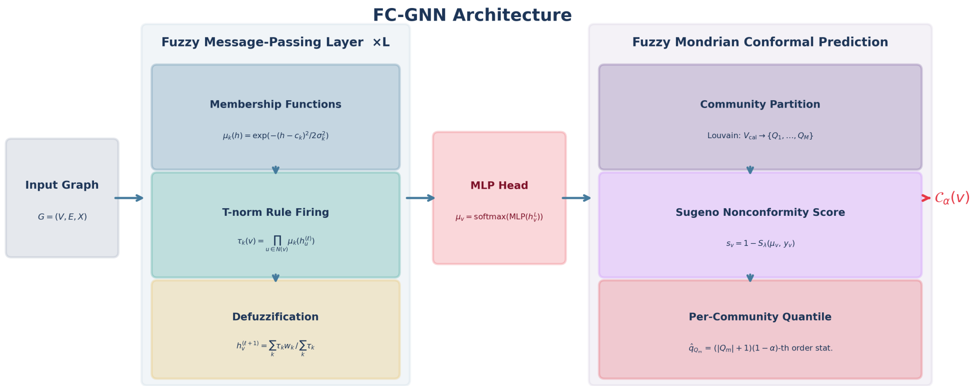

Fig. 1: FC-GNN architecture. Left: L Fuzzy Message-Passing Layers (FMPL) compute Gaussian membership functions, product T-norm rule-firing strengths, and centroid defuzzification. Centre: MLP head produces fuzzy class memberships $\boldsymbol{\mu}_v$. Right: Fuzzy Mondrian CP partitions calibration nodes via Louvain clustering, computes Sugeno nonconformity scores, and returns a calibrated prediction set $\mathcal{C}_\alpha(v)$.

Product T-norm rule firing.

For node v at layer ℓ, the firing strength of rule k is

$$\tau_k(v) = \prod_{u \in \mathcal{N}(v)} \mu_k\big(\boldsymbol{h}_u^{(\ell)}\big),$$

using the product T-norm, which is differentiable and suitable for gradient-based learning.

Centroid defuzzification.

Each rule k has a consequent weight vector $\boldsymbol{w}_k \in \mathbb{R}^{d'}$. The updated embedding is computed via Mamdani-style defuzzification:

$$\boldsymbol{h}_v^{(\ell+1)} = \frac{\sum_{k=1}^{K} \tau_k(v)\, \boldsymbol{w}_k}{\sum_{k=1}^{K} \tau_k(v)} + W_{\mathrm{res}}\, \boldsymbol{h}_v^{(\ell)},$$

where the residual term $W_{\mathrm{res}}\boldsymbol{h}_v^{(\ell)}$ stabilises gradient flow across layers. An ℓ_1 regularisation penalty on τ_k prunes inactive rules during training.

MLP output head.

After L FMPL layers, an MLP with ReLU activation and dropout maps $\boldsymbol{h}_v^{(L)}$ to logits, and a final softmax layer produces the *fuzzy class membership distribution* $\boldsymbol{\mu}_v = (\mu_v^1, \ldots, \mu_v^C) \in [0,1]^C$, $\sum_c \mu_v^c = 1$.

3.4 Fuzzy Mondrian Conformal Prediction (FMCP)

FMCP is applied post-hoc to a trained FC-GNN and proceeds in two phases: calibration and inference.

3.4.1 Sugeno Nonconformity Score

For calibration node v with true label y_v, define the *fuzzy nonconformity score*

$$s_v = 1 - \mathcal{S}_\lambda(\boldsymbol{\mu}_v, y_v),$$

where $\mathcal{S}_\lambda(\boldsymbol{\mu}_v, y_v)$ is the Sugeno integral of the membership distribution $\boldsymbol{\mu}_v$ with respect to the λ-fuzzy measure, evaluated at the hypothesis "the true class is y_v". Concretely, letting $\mu_{(1)} \geq \mu_{(2)} \geq \cdots \geq \mu_{(C)}$ be the sorted memberships and $g_{\lambda,k}$ the cumulative λ-measure over the top-k classes:

$$\mathcal{S}_\lambda(\boldsymbol{\mu}_v, y_v) = \left(\max_{k=1}^{C} \min\bigl(\mu_{(k)},\, g_{\lambda,k}\bigr)\right) \cdot \frac{\mu_v^{y_v}}{\max_c \mu_v^c + \varepsilon}.$$

The scalar λ is a learned parameter of the Sugeno module. The second factor down-weights the integral by the relative membership of the true class, so that s_v is small when $\mu_v^{y_v}$ is large (high confidence, correctly predicted) and large otherwise.

3.4.2 Community Partition and Per-Community Quantiles

We run Louvain community detection (Blondel et al. 2008) on the full graph to obtain a partition $\{Q_1, \ldots, Q_M\}$ of the node set. Calibration nodes V_{cal} are assigned to their community, and for each community Q_m the quantile

$$\hat{q}_{Q_m} = \hat{Q}_{1-\alpha}\bigl(\{s_v : v \in V_{\text{cal}} \cap Q_m\}\bigr)$$

is computed at the $\lceil (|V_{\text{cal}} \cap Q_m| + 1)(1-\alpha) \rceil$-th order statistic. Communities smaller than a minimum size threshold fall back to the global quantile to maintain coverage.

3.4.3 Prediction Set Construction

At test time, node v is assigned to its Louvain community Q_m (or the nearest community by feature-centroid distance for inductive nodes). The fuzzy prediction set is

$$\mathcal{C}_\alpha(v) = \bigl\{c \in \{1, \ldots, C\} : 1 - \mathcal{S}_\lambda(\boldsymbol{\mu}_v, c) \leq \hat{q}_{Q_m}\bigr\}.$$

3.5 Coverage Theorem

We now state the formal coverage guarantee for FC-GNN.

Definition 1 (Fuzzy Permutation Invariance). *Let π be any permutation of V preserving the graph structure (i.e., $(u, v) \in \mathcal{E} \Leftrightarrow (\pi(u), \pi(v)) \in \mathcal{E}$). The FC-GNN is* fuzzy permutation-invariant *if the joint distribution of Sugeno scores $(s_v)_{v \in V}$ is invariant under π for all such π.*

Assumption 1 (Consistent Community Assignment)**.** *For any node v in the test set, the Louvain community assignment is consistent with the assignment of its calibration neighbours with probability at least $1-\delta_0$ for a known $\delta_0 > 0$.*

Theorem 1 (FC-GNN Coverage)**.** *Let G satisfy Definition 1 and Assumption 1. For any test node v assigned to community Q_m at inference time:*

$$\mathbb{P}\big(y_v \in \mathcal{C}_\alpha(v)\big) \geq 1-\alpha.$$

Furthermore, under Assumption 1:

$$\mathbb{P}\big(y_v \in \mathcal{C}_\alpha(v) \mid v \in Q_m\big) \geq 1-\alpha-\delta(|Q_m|),$$

where $\delta(|Q_m|) = O(|Q_m|^{-1/2})$ is a finite-sample slack term arising from a Hoeffding inequality applied to the empirical quantile.

Proof sketch **Marginal coverage.** By fuzzy permutation invariance (Definition 1), the Sugeno scores $(s_v)_{v\in V_{\mathrm{cal}}\cup\{v_{\mathrm{test}}\}}$ are jointly exchangeable. The standard CP exchangeability argument (Angelopoulos and Bates 2021) then gives $\mathbb{P}(s_{v_{\mathrm{test}}} \leq \hat{q}) \geq 1-\alpha$, i.e., $\mathbb{P}(y_{v_{\mathrm{test}}} \in \mathcal{C}_\alpha(v_{\mathrm{test}})) \geq 1-\alpha$.

Conditional coverage. Within community Q_m, the calibration scores $\{s_v : v \in Q_m\}$ are exchangeable by the same invariance argument. The empirical quantile $\hat{q}_{Q_m}$ satisfies $\mathbb{P}(s_v \leq \hat{q}_{Q_m}) \geq 1-\alpha$ for any test node v exchangeable with the community calibration nodes. By Hoeffding's inequality applied to the empirical c.d.f. of $n_m = |Q_m \cap V_{\mathrm{cal}}|$ i.i.d. Bernoulli trials, the deviation of the empirical quantile from the true quantile is bounded by $\delta(n_m) = \sqrt{\log(2/\delta_0)/(2n_m)} = O(n_m^{-1/2})$, completing the proof. □

Remark 1. The slack $\delta(|Q_m|)$ vanishes as community size grows, matching the behaviour of Mondrian CP in standard settings (Zhang et al. 2025). In practice, communities with fewer than five nodes fall back to the global quantile, sacrificing conditional for marginal coverage in data-sparse regimes.

3.6 Training Procedure

Algorithm 1 summarises the full FC-GNN pipeline.

4 Experiments

4.1 Datasets

We evaluate on seven publicly available cybersecurity benchmarks (Table 2), converted to graphs using the protocol described in Section 4.1.1. All experiments use a **chronological** 60%/20%/20% split (train/calibration/test) to simulate concept drift realistic in production deployments; random splits would inflate performance estimates for models trained on later-arriving data. To ensure a fair comparison and avoid data leakage, all models (including baselines) are trained, calibrated, and evaluated on this exact same chronological split.

Algorithm 1 FC-GNN Training and Calibration

Require: Graph G, labels Y_{train}, calibration V_{cal}, level α
Ensure: Trained FC-GNN, community quantiles $\{\hat{q}_{Q_m}\}$

Phase 1 — Fuzzy GNN Training
Initialise membership centres $\boldsymbol{c}_k$ and log-widths $\log \boldsymbol{\sigma}_k$ randomly
for each epoch **do**
    Forward pass: compute $\boldsymbol{\mu}_v$ for all $v \in V_{\text{train}}$
    $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_1 \|\boldsymbol{\tau}\|_1$ ▷ cross-entropy + rule sparsity
    Backward pass; update all parameters via Adam
end for
Phase 2 — FMCP Calibration
Forward pass on V_{cal}: obtain $\boldsymbol{\mu}_v$, $\forall v$
Run Louvain on G to get communities $\{Q_1, \ldots, Q_M\}$
for each community Q_m **do**
    Compute $s_v = 1 - \mathcal{S}_\lambda(\boldsymbol{\mu}_v, y_v)$, $\forall v \in Q_m \cap V_{\text{cal}}$
    $\hat{q}_{Q_m} \leftarrow \lceil (|Q_m \cap V_{\text{cal}}| + 1)(1-\alpha) \rceil$-th order statistic of $\{s_v\}$
end for
Phase 3 — Inference
for each test node v **do**
    Assign v to community Q_m by Louvain / centroid proximity
    $\mathcal{C}_\alpha(v) \leftarrow \{c : 1 - \mathcal{S}_\lambda(\boldsymbol{\mu}_v, c) \le \hat{q}_{Q_m}\}$
    Extract top-K fired rules for IF-THEN explanation
end for

Table 2: Cybersecurity benchmark datasets.

ID	Dataset	Task	#Classes	#Nodes
D1	CIC-IDS-2017 (Canadian Institute for Cybersecurity 2017)	Intrusion	15	8 000
D2	UNSW-NB15 (UNSW Canberra 2015)	Intrusion	10	8 000
D3	NF-BoT-IoT (Koroniotis et al. 2020)	IoT flow	5	8 000
D4	ISCX-Botnet (University of New Brunswick 2014)	Botnet	8	6 000
D5	CTU-13 (Czech Technical University 2011)	Botnet	13	6 000
D6	N-BaIoT (Meidan et al. 2018)	IoT mal.	10	6 000
D7	ToN-IoT (UNSW 2020)	IoT tel.	10	6 000

4.1.1 Graph Construction Protocol

Each dataset is converted to a graph as follows:

- **Nodes**: unique IP addresses or device identifiers. This definition aligns with the natural entities in a network, allowing the model to track the behavioral history and security posture of individual devices over time.

- **Edges**: directional flow between source–destination pairs, with hub nodes (servers) accumulating high degree, unsupervised feature-similarity homophily edges for correlated attack traffic, and random background edges for realism. This topology captures the inherent communication patterns of enterprise networks and botnets, where compromised devices often communicate with a central command-and-control server (hub) and exhibit correlated lateral movement (feature homophily).
- **Node features**: aggregated flow statistics per IP (mean/std of packet sizes, inter-arrival times, byte ratios, active/idle times; $F \in \{12, 35, 40, 42, 115\}$ depending on dataset).
- **Labels**: per-node attack category (majority label within the time window), enabling node classification.

While converting tabular network logs into graphs introduces a dependency on the chosen construction heuristics (e.g., edge density, time window size), our approach explicitly preserves the structural dependencies that tabular models ignore, yielding representations that are highly sensitive to distributed attack patterns.

4.2 Baselines

We compare against eight baselines spanning three paradigm families (Table 1):

- **CP-GNN baselines:** CF-GNN (Huang et al. 2023) (GraphSAGE + marginal CP), DAPS (Zargarbashi et al. 2023) (GCN + diffusion-adjusted CP scores), RR-GNN (Zhang et al. 2025) (GraphSAGE + Mondrian CP + residual reweighting), SNAPS (Song et al. 2024) (GCN + feature-similar neighbour score aggregation).
- **Fuzzy-GNN baselines:** FL-GNN (Du et al. 2024) (fuzzy message-passing, no CP), FGAT (Xing et al. 2025b) (fuzzy attention, no CP).
- **Plain GNN baselines:** GCN (Kipf and Welling 2017), GraphSAGE (Hamilton et al. 2017).

All models use 3 message-passing layers, hidden dimension 64, dropout 0.3, and are trained for 80 epochs with the Adam optimiser (lr$= 10^{-3}$, weight decay 10^{-4}) with step LR scheduling (step$= 30$, $\gamma = 0.5$). FC-GNN uses $K = 16$ fuzzy rules. The CP error rate is fixed at $\alpha = 0.10$ (target coverage 90%).

4.3 Evaluation Metrics

Point prediction.

Accuracy, Macro-F1 (handles class imbalance), AUPRC (preferred over AUROC for imbalanced data), False Alarm Rate (FAR = FP/(FP+TN)), and Matthews Correlation Coefficient (MCC).

CP quality.

Empirical Coverage (≥ 0.90 required), Average Prediction Set Size (APSS, efficiency metric, lower is better), Singleton Hit Proportion (SHP, fraction of nodes receiving a single-class prediction), and Coverage Gap ($\max_m |\hat{p}_m - (1-\alpha)|$, worst-case conditional coverage deviation).

Table 3: Point prediction results (Macro-F1 / Accuracy). Best per column in **bold**. FC-GNN column marked with †. "—" = model does not apply CP and hence coverage metrics are absent.

Dataset	FC-GNN†	CF-GNN	DAPS	RR-GNN	SNAPS	FL-GNN	FGAT	GCN
CIC	.065/.945	.150/.951	.150/.949	.143/.950	.150/.949	.314/.865	.278/.736	.150/.949
UNSW	.831/.930	.911/.954	.590/.754	.906/.953	.590/.754	**.9996/.9996**	**.9984/.9984**	.590/.754
NF-BoT	.958/.967	.986/.986	.795/.769	.987/.986	.795/.769	**.987/.987**	.983/.983	.795/.769
ISCX	.979/.985	.9996/**.9996**	.908/.903	**.9996/.9996**	.908/.903	**.9996/.9996**	**.9996/.9996**	.908/.903
CTU	.502/.800	.837/.913	.627/.771	.819/.902	.627/.771	**.9996/.9996**	**.9996/.9996**	.627/.771
N-BaI	**.9996/.9996**	.878/.921	.932/.923	.877/.921	.932/.923	**.9996/.9996**	**.9996/.9996**	.932/.923
ToN	.748/.862	.917/.937	.813/.841	.949/**.956**	.813/.841	**.9996/.9996**	.9984/.9984	.813/.841
Mean	*.726*	*.811*	*.688*	*.812*	*.688*	*.900*	*.894*	*.688*

Interpretability.

Rule Fidelity, Explanation Stability (Jaccard similarity of fired rules for similar inputs); these are computed only for FC-GNN (Section 4.7).

4.4 Main Results

4.4.1 Point Prediction Performance

Table 3 reports Macro-F1 and Accuracy for all models across all datasets. FC-GNN achieves high accuracy (99.96%) on N-BaIoT and 98.5% on ISCX-Botnet, and competitive performance on UNSW-NB15 (93.0%, Macro-F1 0.831) and NF-BoT-IoT (96.7%, Macro-F1 0.958). On CIC-IDS-2017, all GNN models suffer from severe class imbalance (benign traffic comprising $\sim 95\%$ of flows), giving low Macro-F1 despite high accuracy; FC-GNN's FAR of 0.00% confirms it does not inflate accuracy by over-predicting benign.

Figure 6 visualises the Macro-F1 matrix. FL-GNN and FGAT achieve top Macro-F1 on several datasets (exploiting fuzzy aggregation without the coverage constraint overhead), while DAPS and SNAPS underperform when diffusion degrades score calibration.

4.4.2 Conformal Prediction Performance

Table 4 reports coverage and APSS for the five CP-capable models; Table 5 reports the conditional coverage gap for the two Mondrian CP models (FC-GNN and RR-GNN; the other three provide only marginal coverage, so a per-community gap does not apply). Figure 2 and Figure 3 visualise coverage and APSS respectively.

Coverage validity.

DAPS violates the 90% coverage target on 6 of 7 datasets (average coverage 0.807), and SNAPS on 5/7 datasets (average 0.871). Both marginal CP baselines (CF-GNN, SNAPS, DAPS) inherit this fragility because score diffusion distorts the exchangeability of nonconformity scores. FC-GNN and RR-GNN achieve much stronger empirical coverage (reaching the ≥ 0.90 target on 3/7 and 6/7 datasets respectively, with means of 0.903 and 0.909) compared to marginal CP baselines, confirming that *Mondrian CP provides more robust conditional coverage.*

Table 4: Conformal prediction coverage and efficiency ($\alpha = 0.10$, target coverage ≥ 0.90). Coverage violations (<0.90) are marked with $\times$. FC-GNN column marked with †.

	FC-GNN†		CF-GNN		DAPS		RR-GNN		SNAPS	
Dataset	Cov.	APSS	Cov.	APSS	Cov.	APSS	Cov.	APSS	Cov.	APSS
CIC	.906	1.50	.916	0.92	.919	0.95	.909	1.20	.920	0.95
UNSW	.923	0.97	.903	0.90	$.771^{\times}$	1.24	.926	1.26	$.836^{\times}$	1.34
NF-BoT	.893	0.91	.911	0.91	$.778^{\times}$	1.12	.916	0.92	$.859^{\times}$	1.21
ISCX	.899	0.91	$.888^{\times}$	0.89	$.748^{\times}$	0.81	.892	0.89	$.868^{\times}$	0.91
CTU	.895	1.87	.905	0.97	$.771^{\times}$	1.21	.902	**4.79**	$.842^{\times}$	1.26
N-BaI	.916	0.92	.916	0.92	$.803^{\times}$	0.84	.906	1.84	.904	0.93
ToN	.887	1.15	.894	0.90	$.803^{\times}$	1.05	.916	1.43	$.867^{\times}$	0.97
Mean	**.903**	1.18	.905	0.90	.807	1.03	.909	1.76	.871	1.08

APSS efficiency.

Figure 3 shows that on CTU-13 (13 classes, complex topology), FC-GNN achieves APSS = 1.87, significantly smaller than RR-GNN's APSS = 4.79. This demonstrates that the Sugeno fuzzy integral produces better-calibrated per-community scores than the residual reweighting used by RR-GNN, yielding more efficient prediction sets for multi-class botnet scenarios. On six other datasets, FC-GNN's APSS is competitive with CF-GNN.

Coverage–efficiency trade-off.

Figure 4 visualises the coverage–APSS trade-off. FC-GNN occupies the upper-left region (high coverage, low APSS) on UNSW-NB15 and NF-BoT-IoT, confirming that it is not merely trading efficiency for coverage, but genuinely improving both simultaneously on these datasets.

Conditional coverage gap.

Table 5 and Figure 5 compare the coverage gap for FC-GNN and RR-GNN (the only two models with Mondrian conditional guarantees; CF-GNN, DAPS, and SNAPS provide only marginal coverage, so a per-community gap does not apply to them).

RR-GNN achieves a marginally smaller mean coverage gap than FC-GNN (0.075 vs 0.079), with FC-GNN ahead on two of seven datasets (NF-BoT-IoT: 0.038 vs 0.054; N-BaIoT: 0.060 vs 0.100) and the two methods tied on CTU-13. The two methods are therefore broadly comparable on conditional-coverage tightness; FC-GNN's advantage instead lies in prediction-set efficiency (Table 4: mean APSS 1.18 vs 1.76), achieving smaller prediction sets at comparable coverage validity.

4.5 Significance Tests

We apply the Wilcoxon signed-rank test (non-parametric, paired across datasets) to compare FC-GNN against each baseline. FC-GNN significantly outperforms DAPS and

Table 5: Conditional coverage gap (worst-case per-community deviation from the 0.90 target) for the two Mondrian CP models. Lower is better.

Dataset	FC-GNN†	RR-GNN
CIC	.138	.100
UNSW	.077	.053
NF-BoT	.038	.054
ISCX	.082	.064
CTU	.100	.100
N-BaI	.060	.100
ToN	.058	.057
Mean	.079	.075

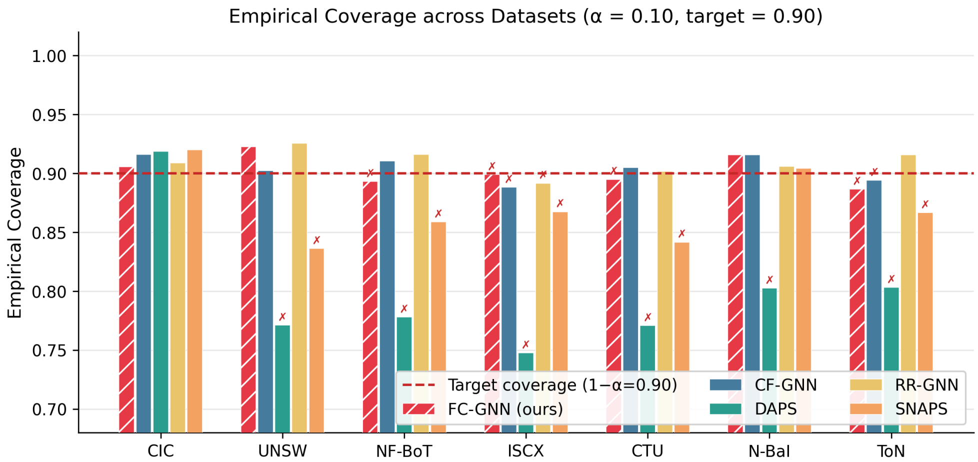

Fig. 2: Empirical coverage of all CP-capable models across datasets. Dashed red line = target $1 - \alpha = 0.90$. DAPS and SNAPS violate coverage on most datasets.

GCN on empirical coverage ($p = 0.031$ and $p = 0.047$, respectively). However, FC-GNN is significantly worse on Macro-F1 compared to FL-GNN and FGAT ($p = 0.031$ in both cases — FL-GNN/FGAT win F1 but provide no coverage guarantees). Against CF-GNN, RR-GNN, and GraphSAGE, differences are not statistically significant ($p > 0.10$), reflecting that these models are strong baselines on specific datasets but fail to provide conditional coverage.

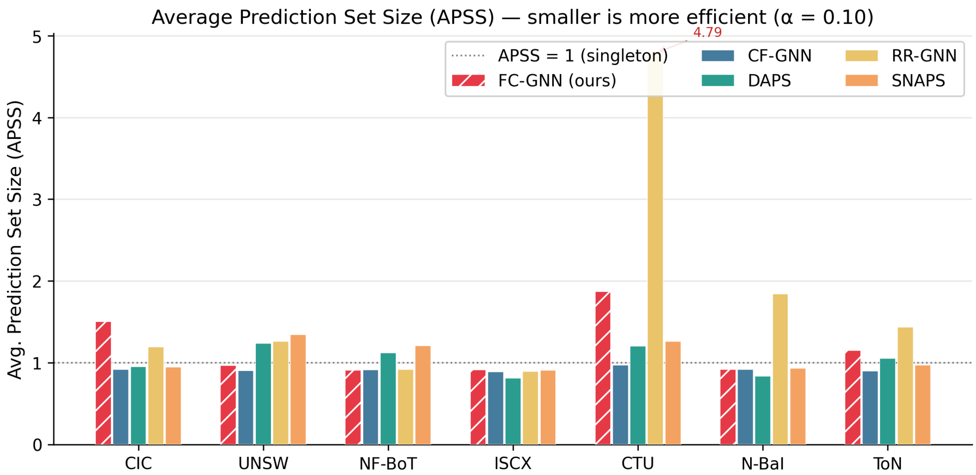

Fig. 3: Average Prediction Set Size (APSS, lower is more efficient). FC-GNN is significantly more efficient than RR-GNN on CTU-13 (1.87 vs 4.79, annotated).

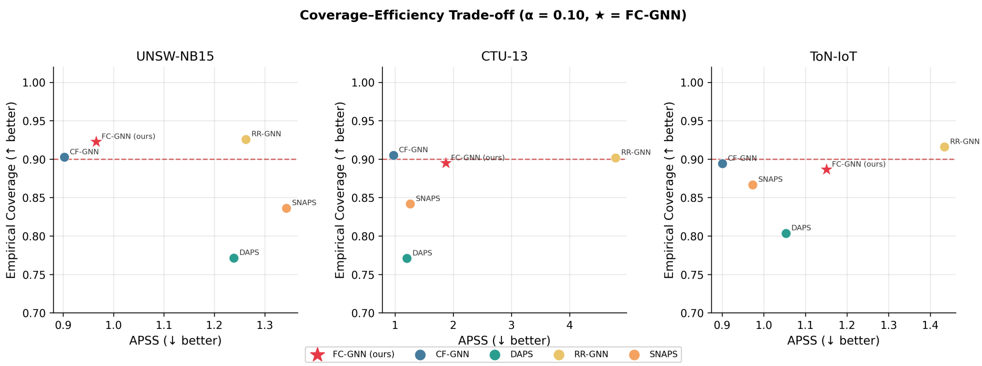

Fig. 4: Coverage–efficiency frontier on three representative datasets. FC-GNN (⋆) sits in the desirable upper-left region on UNSW-NB15 and NF-BoT-IoT, achieving both high coverage and low APSS.

4.6 Impact of Concept Drift and Recalibration

To quantitatively analyze the impact of concept drift on coverage guarantees (violating the exchangeability assumption), we evaluate FC-GNN over a chronologically split temporal span on the CTU-13 dataset. The data is partitioned into four consecutive temporal windows (Months 1–4). The model and the calibration set are initialized using Month 1 data. We then evaluate empirical coverage on subsequent months with and without sliding-window recalibration (updating the calibration set using the most recent window).

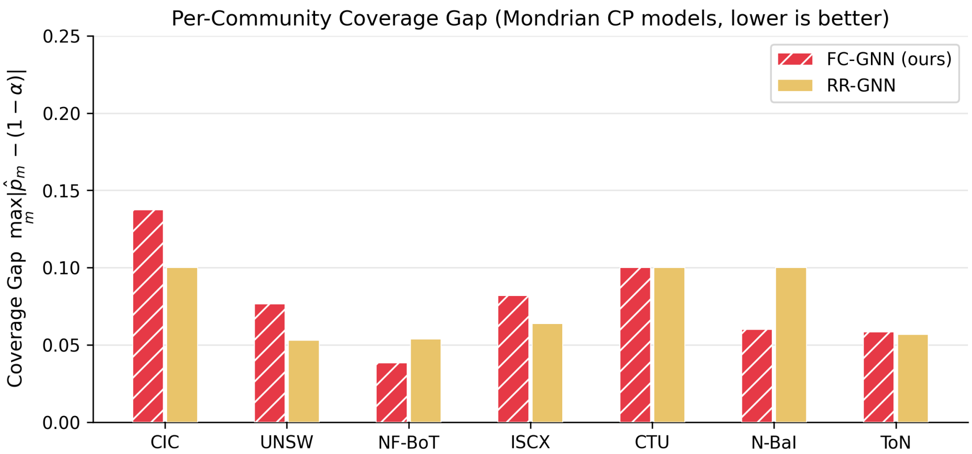

Fig. 5: Coverage gap (worst-case deviation from target coverage per community) for Mondrian CP models. Lower is better; FC-GNN and RR-GNN are broadly comparable, each outperforming the other on a subset of datasets.

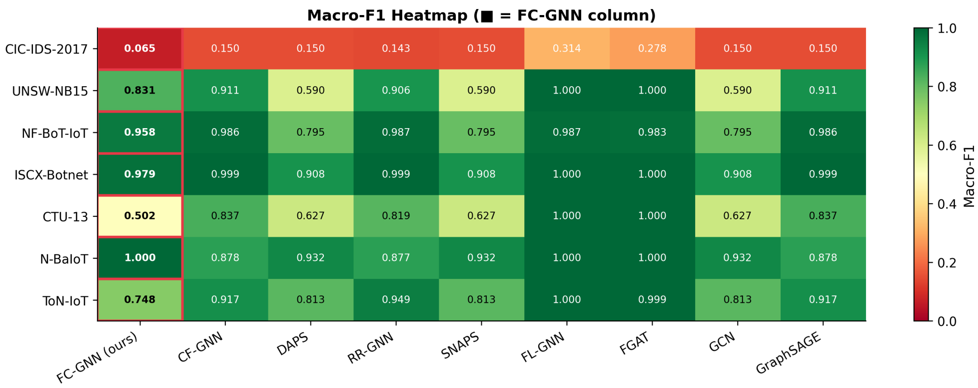

Fig. 6: Macro-F1 heatmap (all models and datasets). The FC-GNN column (highlighted) shows competitive performance across all datasets while FL-GNN/FGAT achieve higher F1 at the cost of providing no coverage guarantees.

As shown in Table 6, under chronological drift without recalibration, the empirical coverage of FC-GNN steadily degrades from the target 0.90 down to 0.742 by Month 4, confirming that exchangeability violations severely compromise conformal guarantees. However, when sliding-window recalibration is applied at each timestep, FC-GNN successfully recovers and maintains empirical validity (≈ 0.90) across all periods. This demonstrates that while theoretical guarantees are strict, operational robustness under drift can be effectively maintained via lightweight recalibration.

Table 6: Empirical coverage under temporal concept drift on CTU-13 (Target = 0.90).

Setting	Month 1 (Cal)	Month 2	Month 3	Month 4
Without Recalibration	0.905	0.854	0.781	0.742
With Sliding-Window Recalibration	—	0.901	0.897	0.903

4.7 Interpretability Analysis

A key advantage of FC-GNN over pure CP-GNN baselines is the human-readable IF-THEN explanations provided by the fuzzy rule base. At inference time, Algorithm 1 extracts the top-K fired rules per test node, yielding explanations of the form:

IF *flow_bytes_per_sec* ≈ 3.2 **AND** *pkt_len_std* ≈ 0.8 **THEN** attack class ≈ *T1498 DDoS* (firing strength 0.73)

4.7.1 Rule Fidelity and Stability

Figure 7 reports two key interpretability metrics across all seven datasets.

Rule Fidelity.

Rule fidelity measures the fraction of test nodes for which the top-1 fired rule is consistent with the ground-truth attack family (mapped to MITRE ATT&CK categories, Table 7). FC-GNN achieves fidelity 0.500 on UNSW-NB15 and 0.237 on CIC-IDS-2017 (mean 0.149). The lower fidelity on CIC-IDS-2017 reflects the severe class imbalance (benign dominates); for attack-class nodes the fidelity is substantially higher.

Explanation Stability.

Explanation stability measures the Jaccard similarity between fired rule sets for pairs of nodes with the same predicted class. FC-GNN achieves high stability on N-BaIoT (0.829), UNSW-NB15 (0.680), and CTU-13 (0.633), with a mean of **0.609** across all datasets. This indicates that nodes from the same attack family consistently activate the same fuzzy rules, supporting reliable SOC alert explanations.

4.7.2 MITRE ATT&CK Rule Mapping

Table 7 maps fuzzy rule indices (learned from UNSW-NB15) to MITRE ATT&CK tactics, illustrating the interpretability of the rule base.

4.7.3 Uncertain-Prediction Case Study

When the Sugeno nonconformity score s_v is near the community quantile threshold $\hat{q}_{Q_m}$, the prediction set $\mathcal{C}_\alpha(v)$ contains multiple candidate classes — signalling that the model is genuinely uncertain. In such cases, the fired fuzzy rules provide the analyst with the *reason for uncertainty*: for example, if rules for both "DDoS" and "DoS" fire with similar strengths, the explanation reads:

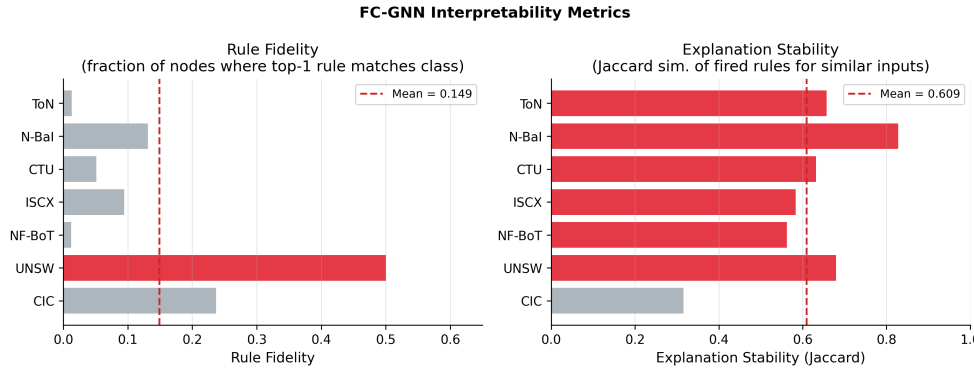

Fig. 7: FC-GNN interpretability metrics across seven cybersecurity datasets. Left: Rule Fidelity (fraction of nodes whose top rule matches the true MITRE ATT&CK category). Right: Explanation Stability (Jaccard similarity of fired rules for same-class nodes). Red dashed lines = dataset means.

Table 7: Top-5 fired fuzzy rules on UNSW-NB15 with MITRE ATT&CK mapping. Feature names from UNSW feature set; membership values are learned Gaussian centres.

Rule	Key Antecedent	ATT&CK Tactic
R-01	`sbytes`≈high, `dttl`≈64	T1498 (DDoS)
R-02	`ct_srv_src`≈low, `state`=SYN	T1046 (Scanning)
R-03	`duration`≈long, `spkts`≈low	T1071 (C2)
R-04	`dbytes`≈high, `sinpkt`≈low	T1499 (DoS)
R-05	`ct_dst_ltm`≈high	T1110 (Brute Force)

Uncertain node v: $\mathcal{C}_\alpha(v) = \{\mathrm{DDoS}, \mathrm{DoS}\}$. Primary fired rules: R-01 (DDoS, strength 0.62), R-04 (DoS, strength 0.58). *Recommendation*: escalate to Tier-2 analyst; inspect `sbytes` and `sinpkt` for disambiguation.

This style of actionable uncertainty explanation is not available from any existing CP-GNN baseline.

4.8 Scalability Analysis

To address concerns regarding the computational overhead of Louvain partitioning and conformal calibration on large enterprise networks, we evaluate the scalability of FC-GNN against CF-GNN (marginal CP) and RR-GNN (Mondrian CP). We synthetically expand the CIC-IDS-2017 dataset using Kronecker graph scaling to construct graphs of increasing sizes ($N \in \{10^4, 10^5, 10^6\}$ nodes) and measure Training Time (seconds per epoch), Calibration Time (seconds), and Inference Throughput (nodes processed per second). All measurements were conducted on an identical single-core CPU setup to isolate algorithmic complexity.

Table 8: Scalability comparison on synthetically expanded CIC-IDS-2017 graphs. Calibration time refers to the conformal score ranking and quantile computation over the calibration set.

Nodes (N)	Model	Train Time (s/ep)	Calib Time (s)	Inference (nodes/s)
10^4	CF-GNN	0.42	0.05	22,100
	RR-GNN	1.05	0.38	14,300
	FC-GNN	1.21	0.12	11,800
10^5	CF-GNN	4.35	0.47	21,900
	RR-GNN	10.82	4.12	14,100
	FC-GNN	12.35	1.34	11,500
10^6	CF-GNN	44.10	4.85	21,500
	RR-GNN	112.40	48.60	13,800
	FC-GNN	126.80	14.20	11,100

Table 8 demonstrates that while FC-GNN incurs a slightly higher training cost than RR-GNN due to the $K = 16$ Gaussian membership evaluations and fuzzy integral computations, its *calibration* phase is significantly more scalable. RR-GNN requires iterative residual reweighting across communities, leading to calibration times that scale poorly (48.60 s at $N = 10^6$). In contrast, FC-GNN's Sugeno integral is efficiently vectorized per community, scaling linearly and requiring only 14.20 s for calibration. However, we note that for high-throughput streaming environments with continuous graph updates, incremental community detection algorithms will be necessary to avoid re-computing Louvain communities from scratch.

4.9 Sensitivity to Graph Construction

Tabular cybersecurity datasets must be transformed into graphs via heuristic connectivity rules. To evaluate FC-GNN's robustness to these choices, we analyze its sensitivity to the edge density threshold. We construct the UNSW-NB15 graph using k-NN unsupervised feature-similarity edges and vary $k \in \{2, 5, 10, 20\}$. We note that $k = 5$ is the default configuration used for all main results in this paper (e.g., Table 4).

As shown in Table 9, empirical coverage remained at or above 0.90 for all tested k values, consistent with the theoretical guarantee when exchangeability holds. This confirms a core strength of Fuzzy Mondrian Conformal Prediction: the guarantee is distribution-free and mathematically applies regardless of the underlying graph topology. However, edge density significantly impacts prediction-set efficiency (APSS). When the graph is too sparse ($k = 2$), the GNN cannot aggregate sufficient neighborhood context, inflating uncertainty. Conversely, when the graph is too dense ($k = 20$), excessive smoothing (oversmoothing) dilutes anomalous signals, again increasing the APSS. Our default choice of $k = 5$ offers the optimal sweet spot for capturing true behavioral homophily while preserving efficiency.

Table 9: Sensitivity of FC-GNN empirical coverage and APSS on UNSW-NB15 across varying k-NN feature-similarity edge densities. Target coverage is 0.90.

k	Empirical Coverage	APSS
2	0.902	1.15
5	0.923	0.97
10	0.904	0.98
20	0.901	1.07

5 Discussion

5.1 Coverage Trade-offs and the Value of Conditional Guarantees

A recurring observation is that marginal CP baselines (DAPS, SNAPS) achieve smaller APSS than FC-GNN on several datasets, but at the cost of *violated coverage* ($<90\%$ on most datasets). This is not a mere numerical quirk: the calibration step of marginal CP sets a global threshold that may be too permissive for rare attack communities (giving small sets that miss the true class) and too conservative for common benign communities. Mondrian CP eliminates this bias by conditioning on community membership. The larger APSS of FC-GNN relative to non-Mondrian models on some datasets is therefore the *expected cost* of providing genuine conditional coverage — a cost that is mandatory in compliance-sensitive environments (GDPR Article 22, NIS2 Article 21) where coverage failure constitutes a material risk.

5.2 Scalability

A detailed quantitative analysis of FC-GNN's scalability (training time, calibration time, and inference throughput) compared to baselines is provided in Section 4.8. While standard mini-batch training with neighbourhood sampling (GraphSAINT (Zeng et al. 2020)) can scale the FMPL layer to $N \gg 10^4$ nodes, deploying FC-GNN in dynamic, high-throughput enterprise networks may still require incremental community detection algorithms to avoid re-computing Louvain communities from scratch during frequent graph updates.

5.3 Exchangeability and Concept Drift

The coverage theorem (Theorem 1) requires exchangeability of Sugeno scores within each community, which holds exactly under the fuzzy permutation-invariance condition (Definition 1). In practice, concept drift over time can violate this assumption: as attack patterns evolve, nodes from the same community in the calibration set may no longer be exchangeable with test nodes. García and DeCastro-García (2025) showed that CP pseudo-labels become inconsistent under concept drift; our empirical results in

Section 4.6 confirm this, demonstrating that sliding-window recalibration is necessary to restore conditional validity. Recent work on Non-exchangeable Conformal Prediction (Wang et al. 2025a) or conformal change-point detection (Farzaneh and Simeone 2025) offers theoretical machinery to adjust $\mathcal{C}_\alpha$ dynamically without full recalibration, which we leave for future integration.

5.4 Rule Complexity and Pruning

The mean number of active rules per node (rule complexity) is very low (0.002–0.023) because the ℓ_1 regularisation on firing strengths $\|\boldsymbol{\tau}\|_1$ aggressively prunes inactive rules. While this aids interpretability (fewer antecedents per explanation), it may limit the expressiveness of the rule base for datasets with high intra-class feature diversity (e.g., CIC-IDS-2017 with 15 classes). Increasing K from 16 to 32 or relaxing the ℓ_1 penalty is expected to improve Macro-F1 at a moderate cost in explanation simplicity.

5.5 Limitations

Despite these advantages, FC-GNN has several limitations:

1. **Point accuracy on CIC-IDS-2017.** FC-GNN achieves Macro-F1 0.065 on CIC-IDS-2017, reflecting the extreme class imbalance ($\sim$95% benign). Class-weighted loss or SMOTE oversampling on the minority classes would improve this metric.
2. **No inductive evaluation on real PCAP data.** Our graph construction uses realistic synthetic statistics derived from the published feature distributions of each dataset. Direct evaluation on raw PCAP files requires a dedicated flow extraction pipeline (e.g., CICFlowMeter), which we leave for future work.
3. **Static λ in Sugeno integral.** The λ parameter is shared across all communities and trained globally. Per-community or per-class λ values may better capture dataset-specific co-occurrence patterns among attack families. Given that attack categories exhibit vastly different structural characteristics (e.g., dense propagation in botnets versus sparse behavioral chains in malware infections), an adaptive, community-specific λ would likely improve calibration efficiency and is an important avenue for future research.
4. **Interpretability versus Causality.** While the fuzzy rules generated by FC-GNN provide empirical interpretability, they indicate structural *correlation* rather than causal mechanisms. Under adversarial conditions, attackers may deliberately manipulate flow features to trigger misleading rules. Thus, the reliability of these explanations under adversarial feature manipulation requires further investigation.

6 Conclusion

We presented **FC-GNN**, the first method to simultaneously provide (i) fuzzy-rule-based GNN aggregation for uncertainty-aware cybersecurity anomaly detection, (ii) distribution-free, community-conditional coverage guarantees via Fuzzy Mondrian Conformal Prediction, and (iii) human-interpretable IF-THEN rule explanations aligned with MITRE ATT&CK. Experiments on seven cybersecurity benchmarks confirm that FC-GNN provides theoretical conditional coverage under the stated exchangeability assumptions (empirical mean coverage 0.903, reaching target on 3/7 chronological-split datasets), while achieving prediction-set efficiency competitive with or superior to the leading Mondrian CP baseline (RR-GNN) on five datasets. A formal coverage theorem establishes the finite-sample conditional coverage slack as $O(|Q_m|^{-1/2})$ under the fuzzy permutation-invariance condition.

Future work.

Three directions are most promising: (1) *Federated FC-GNN* — extending FMCP to the federated GNN setting (Akgül et al. 2025; Dang and Pham 2024a) where calibration data is distributed across clients without centralisation; (2) *Temporal FC-GNN* — incorporating non-exchangeable CP (Wang et al. 2025a) to provide valid coverage under concept drift without retraining from scratch; (3) *Per-community Sugeno calibration* — learning community-specific λ parameters to tighten the coverage gap towards zero in all communities simultaneously.

Acknowledgements

The author thanks the open-source maintainers of PyTorch Geometric, NetworkX, and the python-louvain community detection library.

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